Session 6: CPSA Goals

Slides prepared by UMBC Protocol Analysis Lab

Last edited September 5, 2026

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- **Confidentiality:** No unauthorized party is able to access the information sent. For example, an eavesdropping adversary does not learn where Alice and Bob wish to meet.
- **Agreement:** Both parties agree on some set of data values in the protocol. For example, Alice and Bob agree on a meeting location.

Other common security goals

1. Privacy
2. Anonymity
3. Perfect Forward Secrecy
4. Non-Repudiation

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Confidentiality and Agreement (on the secret) needed.

CPSA Goals

- Protocol goals as mathematical statements
- Find strands that satisfy (or don't satisfy) some set of properties
- May reduce time taken to terminate

Some more tips for reading specifications

- FIND THE SECURITY GOALS
  - Are additional SGs needed?
- Modularize the spec into blocks of messages
  - "In the first set of exchanges"
  - "[party1] preshares [data] with [party2]"
- Ignore nonsense
  - "[protocol] does not consider [niche usecase] because..."
  - "We assume clients have access to an ALU" (structural vs. implementation)
  - "Our conformance program handles..."

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- Let's try modeling it!

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